

RODRIGO GOMES DE ARAÚJO

(+351) 914 574 743 | contact@rodrigoaraujo.pt | rodrigoaraujo.pt | [linkedin.com/in/rodrigoaraujo9](https://www.linkedin.com/in/rodrigoaraujo9) | github.com/rodrigoaraujo9

PROFILE

MSc Computer Science student graduating in **July 2027**, interested in **financial systems** and the reliable and secure software that powers them. Experience in **Rust, Python and PostgreSQL**, with projects spanning a decentralized transaction ledger, an analytical data platform and machine-learning prediction. Thesis work on protocols that are **correct and secure by construction**, useful for financial transaction workflows. Rigorous, reliable and comfortable building systems where correctness matters. Based in Porto and available to relocate internationally or work remote.

EDUCATION

Master's **Sep 2025 – Present (Jul 2027)**
Computer Science *University of Porto (FCUP)*

Bachelor's **Sep 2022 – Jul 2025**
Computer Science and Engineering *University of Porto (FEUP)*

RESEARCH

CaST: Cryptographically-aware Session Types **Jul 2026 – Present (Jul 2027)**
Master's Thesis *FreeST, Haskell, Type Systems, Applied Cryptography*

- Researching how to make communication protocols **correct and secure by construction**, so invalid cryptographic operations are rejected before deployment.
- Developing formal models and compiler support for tracking authorization and cryptographic evidence across nested protocols, with applications to **financial transaction and smart-contract workflows**.

PROJECTS

PoW Blockchain for Decentralized Auctions *Rust, Tokio, LibP2P, Docker, Blake2b, Ed25519*
Data Systems Security · 19/20

- Built a decentralized **public ledger** for auctions, where signed transactions are validated by network participants and recorded in an immutable chain.
- Implemented **proof-of-work consensus**, transaction validation, Merkle-tree block commitments and Kademlia peer discovery.
- Automated multi-node Docker simulations with honest and malicious participants, developing skills in distributed trust, fault resilience and adversarial testing.

Portuguese Elections Analytics Platform *PostgreSQL, PostGIS, Python, SQL, OCaml*
Advanced Topics in Databases · 18/20

- Built an end-to-end analytical platform for Portuguese election results, from **Python ETL** of official Excel data into PostgreSQL to an interactive visual application.
- Designed a **star-schema data warehouse** for votes, turnout and seats, with spatial data and indexes supporting analysis from national to parish level.
- Implemented analytical SQL for **trend, vote-swing, turnout and party-performance analysis**, using roll-ups, CUBE, window functions and geospatial queries.

UFC Fight Outcome Prediction System *Python, pandas, NumPy, scikit-learn, Streamlit*
Artificial Intelligence · 20/20

- Built an end-to-end machine-learning system to predict fight outcomes from historical data, engineering **30+ predictive features** from performance, experience and physical statistics.
- Trained and evaluated **six models**. Logistic Regression, Random Forest, Gradient Boosting, SVM, Neural Network and a soft-voting ensemble, using five-fold cross-validation and explicit overfitting checks.
- Built an interactive Streamlit application with probability estimates, model comparisons, feature analysis and visual reports.

TECHNICAL SKILLS

Programming & Data: Rust, Python, SQL, PostgreSQL, C/C++, Haskell, OCaml, Java
Data & Systems: Tokio, LibP2P, Tonic, pandas, NumPy, scikit-learn, PostGIS, Git, UNIX, Docker, Terraform, OpenMP
Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)

AWARDS

2nd Place in IEEE RetroJam 2025 | IEEE UP Student Branch **Oct 2025**

- Built a game from scratch in 48 hours using Rust and Raylib.

Academic Merit Awards **2020, 2021, 2022**

Participation in RoboCup 2016 | RoboCup Federation **Jun 2016**

1st Place in the National Robotics Championship | Instituto Politécnico de Bragança **May 2016**